

Sense of “world model”	What it is	Examples	Act. cond.	Loop	Metric	VLA-cmp.
<i>As a representation (what the model is)</i>						
latent WM (LWM)	latent-dynamics predictor $z_{t+1} = f(z_t, a_t)$	DreamerV3, TD-MPC2	✓	✓	~	~
action-cond. (WAM)	action-conditioned future-frame predictor	iVideoGPT, GR-2	✓	~	~	~
generative (WFM)	text / image-to-video generator	Sora, Cosmos, Genie	~	✗	✓	✗
<i>As a role (what the model is used for)</i>						
neural simulator	learned environment for policy training / eval	UniSim	✓	✓	~	~
evaluation target	benchmarks that score world models	WorldSimBench, WorldModelBench, EWMBench	--	~	✓	✗

Table 1. The term “world model” spans two orthogonal axes the literature routinely conflates: a *representation* axis (what the model is: latent-state dynamics, action-conditioned video, or generative video, a soft boundary since action-conditioned generators straddle WAM and WFM) and a *role* axis (what it is used for: a neural simulator, or the object a benchmark scores). Columns record whether a sense is **action-conditioned**, is used in a closed **loop**, carries a standard evaluation **metric**, and is readily **VLA-comparable** under one protocol (✓ yes, ~ partial, ✗ no, -- n/a). No single sense is action-conditioned, closed-loop, and VLA-comparable at once, which is why a benchmark that contrasts a world-model policy with a VLA policy per capability (§??) is still missing. Evaluation benchmarks score the world model as an *object*; they are not world models acting as judges.